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## Research Statement

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Vibrant firm dynamics—through firm entry, exit, and growth—are central to resource allocation and economic growth. My research examines how firms enter markets, innovate, and grow; how frictions in information, labor, and product markets shape these decisions; and how firm-level responses translate into aggregate outcomes. Methodologically, I use micro-level data to document empirical patterns in firm behavior and develop quantitative macroeconomic models to identify the underlying mechanisms and assess their aggregate implications. Below, I outline the main themes of my ongoing and future research.

### **Firm Innovation, Organization, and Aggregate Allocation**

In my job market paper, “**Ideas and Firm Dynamics when It Takes Two to Tango**,” joint with Jane Olmstead-Rumsey and Honghao Wang, we study how the organization of innovative activity within firms affects inventor allocation and firm dynamics. Using U.S. patent data, we document a large increase in the importance of research teams and a rise in the concentration of inventors at large firms that substantially exceeds the increase in overall employment concentration.

To explain these trends, we develop a framework of endogenous team formation within firms and embed it in a general-equilibrium model of the inventor labor market with knowledge spillovers across firms. Each inventor draws a bilateral match productivity with every other inventor in the firm and can alternatively innovate alone. As a result, expected inventor productivity increases with the number of coworkers, as a larger pool of potential collaborators raises the likelihood of forming a productive research team. Team production therefore generates endogenous returns to scale in idea production.

We estimate the model separately for 1976–1999 and 2000–2018 in the U.S. economy and find that the returns-to-scale benefits of team production increased substantially over time, potentially reflecting greater knowledge specialization at the individual level and creating more complementarity across team members. The model predicts that this change in idea production increases inventor concentration and average firm size while compressing small firms, consistent with the data. Greater returns to team production also increase aggregate innovative output but endogenously reduce knowledge spillovers. Because firms do not internalize these spillovers when hiring inventors, the equilibrium allocation of inventors across firms is inefficient. We find that the rise in team production amplifies this misallocation by a factor of six, reallocating inventors away from smaller firms that generate disproportionately large knowledge spillovers.

I am also beginning a related project on how firms’ organizational structure evolves over time and shapes subsequent innovation and growth. Using Revelio Labs and USPTO patent data, I plan to study how the hierarchy and composition of firm organization evolve over the firm life cycle, how different organizational structures interact with external labor-market conditions, and how these organizational choices shape subsequent innovation and growth.

### **Firm Dynamics with Learning and Labor Market Frictions**

Another strand of my research studies how incomplete information and learning shape firm dynamics and resource reallocation.

In **“Product Switching and Young Firm Dynamics,”** with Karam Jo, we study how product switching shapes firm growth, selection, and aggregate reallocation. Using detailed firm-product administrative data for the U.S. manufacturing sector, we show that firms frequently reorganize what they produce via switching products and that product switching accounts for a substantial share of firm-level and gross output growth. Product switching is also systematically related to product performance: firms are more likely to drop poorly performing products and retain or add better-performing ones, generating an important margin of within-firm selection. Young firms, however, behave systematically differently from mature firms. They are more likely to retain poorly matched products and to drop products that subsequently prove more productive, suggesting that limited information early in the life cycle distorts product-selection decisions.

To explain these patterns and quantify their aggregate implications, we develop a model of product switching over the firm life cycle in which firms learn about product-specific expertise and adjust their product portfolios accordingly. Firms collect and process information about their products subject to an information-processing capacity that evolves with investment and experience. Limited capacity early in the life cycle generates noisier beliefs and gives rise to the distinctive switching behavior of young firms: they are more likely to retain poorly matched products and drop better-matched ones, slowing the reallocation of resources toward high-performing products. The framework highlights product switching as an important mechanism through which firms reallocate activity toward better products and introduces a novel within-firm margin of resource allocation that complements the standard selection of firms in the market. Quantitatively, greater investment in managerial ability accelerates learning and product selection, improving resource allocation both across and within firms. The within-firm margin generates persistent improvements in firm performance and accounts for more than half of the resulting increase in gross output. We also provide empirical evidence on product switching over the firm life cycle and its relationship with managerial ability that is consistent with the model’s predictions.

A complementary line of this research studies learning from the worker side and its interaction with labor market frictions. In **“Workers’ Job Prospects and Young Firm Dynamics,”** currently under revision at the *Review of Economic Studies*, I study how workers’ uncertainty about their future job prospects affects the ability of young firms to recruit and grow. Because young firms have shorter track records, workers face greater uncertainty about their underlying fundamentals and future employment prospects. I develop a directed-search model with Bayesian learning in which this uncertainty generates systematic wage differences across firms along the lifecycle: high-potential young firms must pay relatively more to attract workers, while low-potential young firms pay relatively less, relative to each of their observationally identical mature counterparts. These wage differentials shift worker allocation from high- toward low-potential young firms, reducing business dynamism and aggregate productivity. Using U.S. Census employer–employee matched data, I find empirical patterns consistent with these predictions across firms and industries.

Building on these insights, my ongoing research **“New Workers for New Ideas: Firm Growth under Labor Market Frictions,”** with Karam Jo, Javier Miranda, and Jiamin Wang, studies how labor market frictions shape the composition of firm innovation and growth using German administrative employer–employee matched data. Expansionary innovation can promote growth through creative destruction and resource reallocation, but it often requires skills and occupations that are not already present within the firm and may therefore depend more heavily on external hiring. We study whether firms recruit for new occupations and skills before entering new technological domains, whether labor-market frictions affect the direction and quality of subsequent innovation, and how the resulting composition of innovation shapes employment, productivity growth, and aggregate welfare. The project also extends my earlier work with Karam Jo on heterogeneous innovation, published in the *Journal of Monetary Economics*.

I have also recently become involved in a preliminary project with the Survey of Business Uncertainty team at the Federal Reserve Bank of Atlanta, together with Nick Bloom and Steve Davis. I am helping build a bridge between the SBU and the Census firm-level data to study how firms' expectations, forecast errors, and uncertainty, measured in different ways, relate to subsequent expansion decisions and resource allocation.

### **Market Structure, Prices, and Inflation**

A third strand of my research studies how market structure and retailer dynamics shape price adjustment and inflation. In **"Geospatial Heterogeneity in Inflation: A Market Concentration Story,"** with Michael A. Navarrete, we use NielsenIQ barcode data to document substantial geographic heterogeneity in food inflation across U.S. markets and show that these differences are systematically related to local retail concentration. To identify the role of market structure in price adjustment, we exploit the bird-flu outbreak as a large, plausibly exogenous cost shock and show that more concentrated markets exhibit larger and more persistent inflation responses. We develop a heterogeneous-firm model with customer capital accumulation that rationalizes these patterns.

Related work with John Haltiwanger, David Johnson, Michael A. Navarrete, Matthew Shapiro, and other coauthors studies the measurement of real sales and inflation using transaction-level barcode data and the implications for standard price-index methodology.

Going forward, I plan to continue developing this research agenda around the forces shaping firm growth, innovation, and resource allocation. I have a Special Sworn Status (SSS) to access massive internal datasets of the U.S. Census Bureau, and I am engaged in several U.S. Census Bureau Federal Statistical Research Data Center (FSRDC) projects. Also, I have access to the Nielsen datasets provided by the Kilts Center at the University of Chicago Booth School of Business and Revelio Labs data. My quantitative modeling and empirical research experience, together with access to these data, will support the continued development of this research agenda.